

whose degree divides 4 are

$$T^2 + T + 1, T^4 + T + 1, T^4 + T^3 + 1, T^4 + T^3 + T^2 + T + 1.$$

But $T^4 + T + 1$ is reciprocal to $T^4 + T^3 + 1$, and $(T+1)^4 + (T+1)^3 + 1 = T^4 + T^3 + T^2 + T + 1$. Hence, it is enough to consider the two polynomials $T^2 + T + 1$ and $T^4 + T + 1$.

If $a^2 + a + 1 = 0$, we proceed as in Example 7.7. Namely, we find that $x := ab + ba$ satisfies $xa = (1 + a)x$. Assuming $W_{2,4,T+1}$, we get $x = 0$ and we are done. Now let $a^4 + a + 1 = 0$. If we define $b_{n+1} := b_n a - a^{2^i} b_n$ for $i = 0, \dots, 4$, then a routine computation, which we omit, shows $b_4 = 0$. Assuming $W_{2,4,2^i}$, we get $b_1 = 0$ and we are done. We will prove the required constructive Wedderburn Theorems in the next section.

8. Constructive Wedderburn Theorems

In this section, we will study more closely the constructive Wedderburn Theorems $W_{p,k,f}$ from Definition 7.2. Recall that they state that $ba = f(a)b$ holds in a p^k -ring with $p = 0$ only when $ba = ab$. Because of Theorem 7.5, it would be very desirable to find equational proofs in general. Here, we will just cover some partial results and special cases. We also indicate how the general case could theoretically be handled by a computer program. Even though we already saw in Theorem 7.9 that it is enough to consider $f = T^{p^m}$ for $0 \leq m < k$, we will consider arbitrary polynomials $f \in \mathbb{F}_p[T]$ for the moment.

Lemma 8.1. *If $f \in \mathbb{F}_p[T]$ is constant, then there is an equational proof of $W_{p,k,f}$.*

Proof. In a p^k -ring A with $p = 0$, assume $ba = ub$ holds for some $u \in \mathbb{F}_p$. Then, since u and b commute, we have $b(a - u) = 0$. Since A is reduced, this implies $(a - u)b = 0$, i.e. $ab = ub$. Thus, $ab = ba$. $\square$

For a polynomial f in one variable, we denote by $f^{\circ j}$ the j -fold composition of f with itself.

Lemma 8.2. *If $f \in \mathbb{Z}[T]$ is a polynomial and the equation $ba = f(a)b$ holds in a ring, then for all $i, j \in \mathbb{N}$ we have*

$$b^j a^i = f^{\circ j}(a)^i b^j.$$

More generally, for every $g \in \mathbb{Z}[T]$ we have

$$b^j g(a) = g(f^{\circ j}(a)) b^j.$$

Proof. The second equation follows from the first. The proof of the first is routine, so a sketch should be enough. One first proves the case $j = 1$ by induction on i . Then one proves the case $i = 1$ by induction on j . Then one proves the general case by induction on i . $\square$

Lemma 8.3. *For $u \in \mathbb{F}_p$ there is an equational proof of $W_{p,k,T+u}$.*

Proof. Assume that $ba = (a + u)b$ holds in a p^k -ring with $p = 0$. By Lemma 8.2, we get

$$ba = b^{p^k} a = (a + up^k) b^{p^k} = ab. \quad \square$$

Lemma 8.4. *To give an equational proof of $W_{p,k,f}$, one may assume that a and b are units.*

Proof. Let $ba = f(a)b$. Assume that there is an equational proof of $ba = ab$ in case b is a unit. By Lemma 2.17 we can write $b = ue$, where u is a unit and $e = b^{p^k-1}$ is idempotent, hence central. It suffices to prove that b and ea commute, since b commutes with $(1 - e)a$ anyway (the products vanish). Consider the p^k -ring eA with multiplicative identity e and the homomorphism $\alpha : A \rightarrow eA$, $x \mapsto ex$. Then we get $\alpha(b)\alpha(a) = f(\alpha(a))\alpha(b)$ in eA . If $W_{p,k,f}$ has been proven in eA , where b is a unit, then $\alpha(b) = b$ commutes with $\alpha(a) = ea$ and we are done. We need to make this argument equational, though, and this works as in the proof of Theorem 7.5. Observe that e is neutral for both ea and b . Any proof of $W_{p,k,f}$ can be applied

to the elements ea and b and the multiplicative “pseudo-identity” e . Thus, b and ea commute, and we are done. The reduction to the case that a is a unit works exactly the same. $\square$

When b is a unit, then $W_{p,k,f}$ states that $ba = f(a)b$ implies $f(a) = a$. Thus, we need to show that a is a fixed point of f . We will see in a moment that at least a is a fixed point of a certain power $f^{\circ d}$.

Definition 8.5. Let $f, g \in \mathbb{F}_p[T]$ and $k \geq 1$. We write $f \equiv_k g$ or just $f \equiv g$ (when k is clear from the context) when there is an equational proof of $f(a) = g(a)$ for all elements a of a p^k -ring with $p = 0$. The set of equivalence classes becomes a monoid with respect to the composition of polynomials:

$$(f, g) \mapsto f \circ g.$$

Notice that it is well-defined, and the multiplicative identity is T . The underlying set can be identified with the one of $\mathbb{F}_p[T]/\langle T^{p^k} - T \rangle$, since $f \equiv g$ holds if and only if $T^{p^k} - T \mid f - g$. In particular, this monoid is finite, it has exactly p^{p^k} elements.

Remark 8.6. If f is an element of a finite monoid, whose multiplication we write as $\circ$, then there must be a pair of numbers $i \geq 0$ and $\pi > 0$ such that $f^{\circ i} = f^{\circ(i+\pi)}$. Then the powers of f are eventually periodic:

$$1, f, \dots, f^{\circ i}, f^{\circ(i+1)}, \dots, f^{\circ(i+\pi-1)}, f^{\circ i}, f^{\circ(i+1)}, \dots.$$

We say that f is of *period* π and *index* i . If π is minimal, we call π *the* period, similarly for the index.

In particular, every $f \in \mathbb{F}_p[T]$ has some period and index with respect to the finite monoid $(\mathbb{F}_p[T]/\langle T^{p^k} - T \rangle, \circ, T)$ from Definition 8.5. Both can be efficiently computed for every specific choice of p, k, f because of Lemma 2.9. See also the following example.

Example 8.7.

- (1) Constant polynomials are of period 1 and index 1.
- (2) The polynomial $-T$ is of period 2 and index 0 (also of period 1 when $p = 2$).
- (3) For $p = 2, k = 3, f = T^2 + T$, we compute $f^{\circ 2} = T^4 + T, f^{\circ 3} = T^4 + T^2$ and $f^{\circ 4} = T^8 + T^2 \equiv T^2 + T$, so that f is of period 3 and index 1.
- (4) For $p = 2, k = 4, f = T^2 + 1$, we compute $f^{\circ 2} = (T^2 + 1)^2 + 1 = T^4, f^{\circ 3} = T^8 + 1$ and $f^{\circ 4} = T^{16} \equiv T$, so that f is of period 4 and index 0.

Proposition 8.8. Let $f \in \mathbb{F}_p[T]$ be a polynomial of period $\pi > 0$. Assume that a, b are two elements of a p^k -ring with $p = 0$ with $ba = f(a)b$, and b is a unit. Then we have $f^{\circ d}(a) = a$, where $d := \gcd(\pi, p^k - 1)$. In particular, if $d = 1$, there is an equational proof of $W_{p,k,f}$.

Notice that this result generalizes both Lemma 8.1 and Lemma 8.3, and that the assumption that b is a unit is harmless because of Lemma 8.4. Without this assumption, we can only conclude that b commutes with a^d .

Proof. Choose $i \geq 0$ with $f^{\circ i} \equiv f^{\circ(i+\pi)}$. Lemma 8.2 allows us to compute

$$b^i b^\pi a = b^{i+\pi} a = f^{\circ(i+\pi)}(a) b^{i+\pi} = f^{\circ i}(a) b^i b^\pi = b^i a b^\pi.$$

Since b^i is a unit, this shows that a and b^π commute. But then

$$f^{\circ \pi}(a) b^\pi = b^\pi a = a b^\pi.$$

Since b^π is a unit, we get

$$f^{\circ \pi}(a) = a. \tag{20}$$

Next, we compute

$$f(a)b = ba = b^{p^k} a = f^{\circ p^k}(a) b^{p^k} = f^{\circ p^k}(a) b.$$

Since b is a unit, we get $f(a) = f^{\circ p^k}(a)$. Apply $f^{\circ(\pi-1)}$ to both sides and use $a = f^{\circ\pi}(a)$. Then we get

$$f^{\circ(p^k-1)}(a) = a. \quad (21)$$

If $d := \gcd(\pi, p^k - 1)$, then (20), (21) and the Bézout identity imply $f^{\circ d}(a) = a$. $\square$

Lemma 8.9. *Let $m \in \mathbb{N}$ and let π be the order of $[m]$ in the additive group $\mathbb{Z}/k\mathbb{Z}$. Then the monomial T^{p^m} is of period π and index 0.*

Proof. Let $f := T^{p^m}$. Write $\pi m = sk$ with $s \in \mathbb{N}$. Then $f^{\circ\pi} = T^{p^{m\pi}} = T^{p^{sk}} \equiv T$. $\square$

Theorem 8.10. *Let $p \in \mathbb{P}$ and $k \geq 1$. If $\gcd(k, p^k - 1) = 1$, then there is an equational proof that every p^k -ring with $p = 0$ is commutative.*

Proof. This follows from Theorem 7.9, Proposition 8.8 and Lemma 8.9. $\square$

Remark 8.11. For the lack of a better name, let us call $k \geq 1$ *good* for $p \in \mathbb{P}$ if we have

$$\gcd(k, p^k - 1) = 1,$$

otherwise *bad* for p . With Theorem 8.10 we have solved the equational commutativity problem for good exponents, so let us analyze how many numbers are good. For $p = 2$, the 22 bad numbers $k \leq 100$ are

$$6, 12, 18, 20, 21, 24, 30, 36, 40, 42, 48, 54, 60, 63, 66, 72, 78, 80, 84, 90, 96, 100.$$

For the other 78 numbers $k \leq 100$, we thus have an equational proof that 2^k -rings are commutative. As before, this then leads to many more cases: By Theorem 4.1, every 94-ring is a $2^{\text{lcm}(1,2,5)} = 2^{10}$ -ring, hence commutative. For $p > 2$, every good number must be odd. For $p = 3$ and $p = 5$, most odd numbers are also good. For example, the only odd numbers $k \leq 100$ that are bad for 3 are 39, 55, and the only odd numbers $k \leq 100$ that are bad for 5 are 55, 93. But for $p > 5$ there are much more bad odd numbers. If $k \mid k'$ and k is bad, then k' is bad as well, since $\gcd(k, p^k - 1) \mid \gcd(k', p^{k'} - 1)$.

The following characterization of bad numbers has been found by Max Alekseyev.

Lemma 8.12. *Let $p \in \mathbb{P}$. The set of bad numbers k for p is equal to the union*

$$\bigcup_{q \in \mathbb{P}, q \neq p} q \cdot \text{ord}_q(p) \cdot \mathbb{N}^+.$$

Here, ord_q refers to the order in the group $(\mathbb{Z}/q\mathbb{Z})^\times$.

Proof. Let $k \geq 1$. If k belongs to the union, there is a prime $q \neq p$ with $q \cdot \text{ord}_q(p) \mid k$. Then $q \mid k$ and $q \mid p^k - 1$, so that $\gcd(k, p^k - 1) > 1$. Conversely, if $\gcd(k, p^k - 1) > 1$, there is a prime q with $q \mid k$ and $q \mid p^k - 1$. The second condition implies $q \neq p$ and then becomes equivalent to $\text{ord}_q(p) \mid k$. Since the order divides $q - 1$, it is coprime to q . Hence, $q \cdot \text{ord}_q(p) \mid k$. $\square$

With Theorem 8.10, 2^6 and 3^4 are the only prime powers ≤ 100 for which we don't have an equational commutativity proof yet, see also Table 1. For the moment, the general case seems to be out of reach. But it turns out that $W_{p,k,f}$ can be verified by a computer program, at least theoretically, since it is a *finite* problem:

Proposition 8.13. *There is a finite $\mathbb{F}_p$ -algebra $U_{p,k,f}$ with a concrete finite presentation such that $U_{p,k,f}$ is commutative if and only if $W_{p,k,f}$ holds.*

Proof. Because of Lemma 8.4, it suffices to work with units. (This is not necessary for the proof, but it will decrease the dimension of the algebra below.) We define the $\mathbb{F}_p$ -algebra by the finite presentation $V_{p,k,f} := \mathbb{F}_p\langle X, Y : X^{p^k-1} = 1, Y^{p^k-1} = 1, YX = f(X)Y \rangle$. We saw in